Curriculum Vitae of Giordano De Marzo

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ORCID:	0000-0002-3127-5336
Google Scholar metrics:	h-index: 15, i10-index: 21, citations: 718 (October 2026)
Nationality:	Italian
Date of birth:	24 June 1996

Research profile

I study the collective behavior of AI agents: what populations of language-model agents do when they interact, which laws their collective dynamics follow, and how they compare with human groups. My work combines experimental paradigms from the behavioral sciences with the quantitative theory of complex systems to link the dispositions of individual agents to the behavior of their populations, from a universal law of coordination in AI agent societies (*Science Advances*, 2026) to the conditions under which conformity turns aligned agents into misaligned collectives. Trained as a theoretical physicist (PhD, Sapienza, 2024), I now lead a line of research on collective AI behavior at the Social Data Science Lab in the University of Konstanz, where I have supervised eight completed or submitted master's theses, seven of them on AI agents, with four more ongoing; I also teach in the Social and Economic Data Science master's program.

Academic Positions

02/2024–present	**Postdoctoral Researcher and Lecturer** (fixed-term contract until 01/2030)
Institution	Social Data Science Lab, University of Konstanz
01/2023–01/2026	**Junior Research Fellow**, Complexity Science Hub Vienna (non-resident fellowship)
03/2022–present	**Fellow**, Centro Ricerche Enrico Fermi (CREF), Rome (non-resident affiliation)
02/2021–01/2024	**PhD Fellow**, Sapienza School for Advanced Studies

Education and training

11/2020–03/2024	**PhD in Physics**, Summa cum Laude
Thesis Title	Modeling Complex Digital Systems
Supervisors	Vittorio Loreto, Luciano Pietronero
Institution	Sapienza University of Rome and Enrico Fermi Research Center
09/2018–07/2020	**Master's Degree in Theoretical Physics**, 110/110 with honors
Principal subjects	Statistical Mechanics, Complex Systems
Supervisor	Luciano Pietronero
Institution	Sapienza University of Rome
09/2015–09/2018	**Bachelor's Degree in Physics**, 110/110 with honors
Supervisor	Luciano Pietronero
Institution	Sapienza University of Rome
10/2020–03/2021	**Advanced School in Artificial Intelligence**
Institution	Institute for Cognitive Sciences and Technologies (ISTC-CNR), postgraduate school

Supervision

Master's theses at the University of Konstanz (first supervisor; Social and Economic Data Science). Since 2025 the theses I supervise form a coordinated line of work on the collective behavior of AI agents (marked •), from the social dispositions of individual agents, measured with paradigms from social and cognitive psychology, to the dynamics of agent populations; seven have been completed or submitted, and their results serve as pilot studies for the research program proposed here. One thesis has been accepted at EMNLP 2026 and nominated for its Outstanding Paper Award, and another has become a preprint. In the summer semester of 2026 I led a team of five students and research assistants within the lab.

2026 • **Hanna Sophie Müller**, From Social Psychology to Artificial Intelligence: An Investigation of the Bystander Effect in Large Language Models (with W. Gaißmaier).

2026 • **Fabian Mahner**, Artificial Conformity: How Peer Pressure Can Undermine the Wisdom of the Crowd in Vision Language Models.

2026 • **Lorenz Rückert**, Coordination and Conformity in Multi-Option LLM Systems.

2026 • **Miranda Barros Everet**, Testing Conformity in Large Language Models Using Simulated Social Media Posts (with W. Gaißmaier).

2026 • **Katrina Jones**, Conjoint Auditing: A Framework for Measuring the Social Impact of Large Language Model Deployment. Accepted at EMNLP 2026 (to appear in the ACL Anthology); nominated for the Outstanding Paper Award.

2025/26 • **Jakob Aman**, Cooperation, Competition and Collapse: How Model Scale, Group Size and Communication Shape Strategic Behavior of AI Agents in Collective Risk Social Dilemmas.

2025/26 • **Adrian Ruhe**, Locating and Manipulating Representations of Information Quality in Large Language Models.

2025 **Raphaela Keßler**, Graph Neural Networks for News Domain Classification on Telegram. Published as a preprint (arXiv:2608.02399).

Ongoing • **Jonas Schrade**, Does Social Feedback Cause Structural Polarization Among LLM Agents?; • **Samuel Rauh**, Multi-Agent Systems for Universal Generative Urban Planning; **Maite Küster**, diffusion-based downscaling for urban climate analysis (with the EU project FOCAL); **Ludwig Kunz**, network analysis of lobby meetings in the European Parliament.

2021–2022 **External supervisor of three master's theses at Sapienza University of Rome** (A. Bellina, F. Pandolfelli, G. Iannelli; with V. Loreto, L. Pietronero, C. Castellano). Two of the three are now postdocs and co-authors.

Technical leadership

03/2025–Ongoing **Co-founder**, energ.IA. Unpaid technical advisory role, alongside my academic position, in a startup that uses data science and AI to automate the design of renewable-energy facilities (product: Ferward). I set the technical direction, from backend and cloud architecture (Azure) to frontend, and oversee a development team of five (backend developer, AI and cloud apprentice, tester, frontend consultant, project-management consultant).

Teaching

All courses at the Master's Degree in Social and Economic Data Science, University of Konstanz, as lecturer.

2024/25, 2025/26 **Network Science of Socio-Economic Systems** (course I designed)

2024, 2025 **Deep Learning for the Social Sciences** (course I designed)

2024 **Computational Modeling of Social Systems**

2023 **Classical Mechanics**, teaching assistant, Bachelor's Degree in Physics, Sapienza University of Rome

Software and data

All code and data from my research are released openly.

2026 **Moltbook crawler and dataset**: continuously updated record of an agent-only social platform (369k posts, 3M comments at first release). HuggingFace: huggingface.co/datasets/giordano-dm/moltbook-crawl; code: github.com/giordano-demarzo/moltbook-api-crawler

2026 **Opinion dynamics with AI agents**: experimental framework for collective behavior of LLM populations (nine models, 100 opinion pairs). github.com/giordano-demarzo/Opinion-Dynamics-with-AI-Agents

2024–2026 **LLMs-Opinion-Dynamics**: code and data of the *Science Advances* paper, archived on Zenodo (doi:10.5281/zenodo.17911594). github.com/giordano-demarzo/LLMs-Opinion-Dynamics

Research visits and consulting

03/2023–06/2023 **Visiting PhD Student**, University of Konstanz. Emergent and collective behavior in groups of large language models; network growth with generative agents.

10/2022–12/2022 **Visiting PhD Student**, Sony CSL Paris. Effect of recommendation algorithms on polarization and novelty discovery in groups of agents.

11/2021–05/2022 **Junior Data Scientist**, Complexity Science Hub Vienna. Reputation algorithms for collaborative systems.

2021–2024 **External consultant**, International Labour Organization (three contracts) and United Nations University. NLP analysis of large online job-vacancy datasets; economic-complexity analysis of national labour markets.

08/2020–12/2023 **External consultant**, Translated. Language economics: algorithms to identify the best languages for online sales and forecast market opportunities.

Selected publications

The three publications submitted with this application. The complete list of publications is provided as a separate document.

2026 De Marzo, Giordano, Claudio Castellano, and David Garcia. "AI agents can coordinate via majority-following beyond human scale." *Science Advances* 12, eaea6091 (2026).

2023 Bellina, Alessandro, Claudio Castellano, Paul Pineau, Giulio Iannelli, and Giordano De Marzo. "Effect of collaborative-filtering-based recommendation algorithms on opinion polarization." *Physical Review E* 108, 054304 (2023). Contribution: conceived the study and the model, supervised the junior researchers who carried out the work (a former master's student of mine and a research intern), and led the writing of the manuscript; senior author.

2020 De Marzo, Giordano, Andrea Zaccaria, and Claudio Castellano. "Emergence of polarization in a voter model with personalized information." *Physical Review Research* 2, 043117 (2020).

Further work on AI agent societies: the first study of how networks form among LLM agents (arXiv:2312.06619, 2023); collective misalignment through conformity (arXiv:2605.10721, in revision at *Nature Communications*); conformity and social impact on AI agents (arXiv:2601.05384); collective behavior on the agent-only platform Moltbook (arXiv:2602.09270); copying as the mechanism behind agents' collective behavior in the wild (arXiv:2609.09150); the Collective Turing Test (*Scientific Reports*, 2026).

Invited Talks

27/09/2026 **Keynote**, First Workshop on Sovereign AI for Collaborative and Pluralistic AI Ecosystems, ACM HCOMP 2026 (Washington, DC). *Emergent behaviours in AI agent societies.*

27/09/2026 **Invited talk**, The Future of Research in the Age of Human–AI Collective Intelligence, ACM Collective Intelligence Conference 2026 (Washington, DC). *Collective behavior and emergent coordination in AI agent societies.*

2025 **Lecture**, Summer School of the Centre for the Advanced Study of Collective Behaviour (University of Konstanz and Max Planck Institute of Animal Behavior).

12/06/2025 GenAI Research Retreat 2025, invitation-only retreat on generative AI for complex social systems (Brussels, Belgium). *Emergent behaviors in AI agent societies.*

20/03/2025 **Main invited talk (Hauptvortrag)**, Focus Session on Large Language Models, Social Dynamics, and Assessment of Complex Systems, DPG Spring Meeting (Regensburg, Germany). *Emergent behaviors in LLM-populated societies.*

27/05/2024 Social Simulation with Large Language Models, invitation-only workshop (Complexity Science Hub, Vienna, Austria). *Emergent behaviors in LLM-populated societies.*

Invited Seminars

2026 Universidade de Fortaleza (Unifor), Brazil

2025 Department of Network and Data Science, Central European University, Research Seminar Series

2024 Complexity Science Hub Vienna, CSH Talk Series

2024 MPI for Dynamics and Self-Organization, Neural Systems Theory Group

Conference Contributions

Contributed talks: Conference on Complex Systems CCS'25 (Siena, 2025); Conference on Complex Systems CCS'24 (Exeter, 2024); NetSci 2024 (Quebec City); NetSciX 2024 (Venice); NetSci 2023 (Vienna); Complex Networks and their Applications 2022 (Palermo); Conference on Complex Systems 2022 (Palma de Mallorca); Complex Networks 2022 (Lipari); I Conference of the Italian Society of Statistical Physics (Parma, 2021). **Posters:** NetSci 2023 (Vienna); II Conference of the Italian Society of Statistical Physics (Parma, 2022).

Scholarships, Grants and Awards

2024 **EPO CodeFest**, second classified team (10,000€)

2023 **Research Credits**, OpenAI (5,000€)

2023 **Research Grant**, DAAD (4,000€)

2022 **Research Grant**, Imminent (20,000€)

2022 **Research Grant**, Fondazione Zegna (5,000€)

2020–2023 **Scholarship**, Sapienza School for Advanced Studies (3 years of accommodation)

2016–2020 **Scholarship**, Sapienza University (8,000€)

2017–2019 **Working Scholarship**, Sapienza University (2,000€)

2018 **Degree Award**, Sapienza University (1,200€)

2015 **Seventh Place**, Italian Natural Science Olympiad; **Bronze Medal**, Italian Physics Olympiad

Academic Service

2025–Ongoing **Member**, Examination Board, Master's Degree in Social and Economic Data Science (SEDS), University of Konstanz

2024–2025 **Organizer**, CDM Colloquium Series, Department of Politics and Public Administration, University of Konstanz

Peer reviewer for *Nature, Nature Communications, Nature Physics, Nature Machine Intelligence, Neural Networks, Scientific Reports, Physica A, EPJ B, EPJ Data Science, Online Social Networks and Media, Monthly Notices of the Royal Astronomical Society, Applied Network Science.*

2023–2025 **Organizer** of four young-researcher workshops: Fostering Research Synergies (IMT Lucca, 2023 and 2025; CREF Rome, 2024) and the PhD and Young Researchers Workshop (CSH Vienna, 2024)

Media Coverage

08/2026 *ScienceAlert*: "1,000 AI Agents Started Agreeing Without Anyone Telling Them To". Interview on the *Science Advances* study.

08/2026 *PsyPost*: "Artificial intelligence agents spontaneously conform to the majority opinion". Interview.

08/2026 *Il Fatto Quotidiano*: "Gli agenti di IA si mettono d'accordo da soli: seguono la maggioranza anche in gruppi di mille" (in Italian). Coverage of the *Science Advances* study.

05/2026 *Epsiloon*: "De quoi parlent les IA entre elles?" (in French). Interview on Moltbook.

03/2026 *Nature* (News): "The first 'AI societies' are taking shape: how human-like are they?". Interview; features the Moltbook study.

01/2026 *New Scientist*: "Chess can be made fairer by rearranging the pieces". Expert comment.

07/2025	*Der Standard*: “Artensterben mit Methoden der Wirtschaftswissenschaften berechnet” (in German). Coverage of the food-web study.
10/2024	*New Scientist*: “AIs can work together in much larger groups than humans ever could”. Interview, with comment by Robin Dunbar.
04/2023	*Wired Italia*: “La scienza ha creato una nuova classificazione delle aperture degli scacchi” (in Italian). Coverage of the chess-openings study.